Enable NAS storage

ONTAP 9

NetApp
March 27, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap/task_nas_enable_linux_nfs.html on March 27, 2024. Always check docs.netapp.com for the latest.

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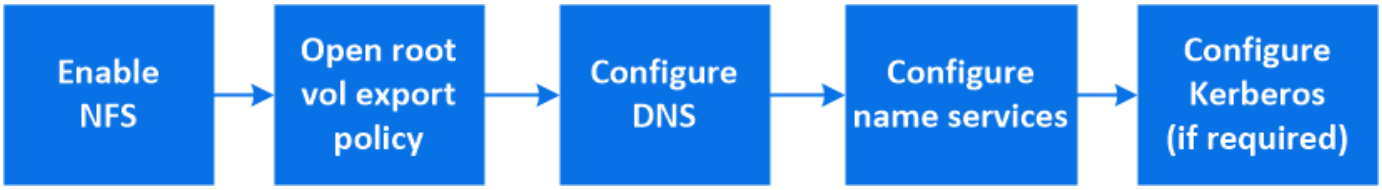
Enable NAS storage

Enable NAS storage for Linux servers using NFS

Create or modify storage VMs to enable NFS servers for serving data to Linux clients.

This procedure enables a new or existing storage VM for the NFS protocol. It is assumed that configuration details are available for any networking, authentication, or security services required in your environment.



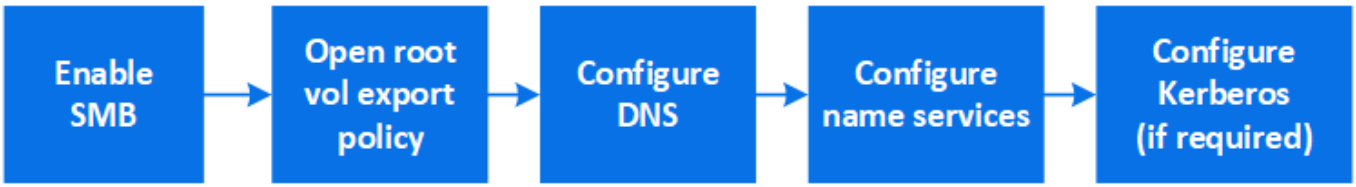
Steps

1. Enable NFS on a storage VM.
   a. For new storage VMs: click **Storage > Storage VMs**, click **Add**, enter a storage VM name, and in the **SMB/CIFS, NFS, S3** tab, select **Enable NFS**.
      - Confirm the default language.
      - Add network interfaces.
      - Update storage VM administrator account information (optional).
   b. For existing storage VMs: click **Storage > Storage VMs**, select a storage VM, click **Settings**, and then click ⚙ under **NFS**.
2. Open the export policy of the storage VM root volume:
   a. Click **Storage > Volumes**, select the root volume of the storage VM (which by default is *volume-name* _root), and then click on the policy that is displayed under **Export Policy**.
   b. Click **Add** to add a rule.
      - Client specification = `0.0.0.0/0`
      - Access protocols = NFS
      - Access details = UNIX Read-Only
3. Configure DNS for host-name resolution: click **Storage > Storage VMs**, select the storage VM, click **Settings**, and then click ⚙ under **DNS**.
4. Configure name services as required.
   a. Click **Storage > Storage VMs**, select the storage VM, click **Settings**, and then click for ⚙ LDAP or NIS.
   b. Include any changes in the name services switch file: click ✎ in the Name Services Switch tile.
5. Configure Kerberos if required:
   a. Click **Storage > Storage VMs**, select the storage VM, and then click **Settings**.
   b. Click → in the Kerberos tile and then click **Add**.

Enable NAS storage for Windows servers using SMB

Create or modify storage VMs to enable SMB servers for serving data to Windows clients.

This procedure enables a new or existing storage VM for the SMB protocol. It is assumed that configuration details are available for any networking, authentication, or security services required in your environment.



Steps

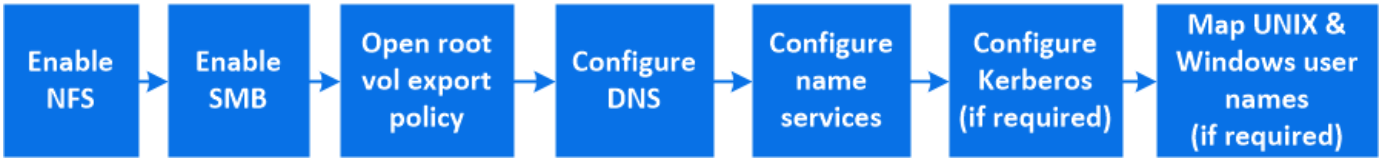
1. Enable SMB on a storage VM.
   a. For new storage VMs: click **Storage > Storage VMs**, click **Add**, enter a storage VM name, and in the **SMB/CIFS, NFS, S3** tab, select **Enable SMB/CIFS**.
      - Enter the following information:
        - Administrator name and password
        - Server name
        - Active directory domain
      - Confirm the Organizational Unit.
      - Confirm the DNS values.
      - Confirm the default language.
      - Add network interfaces.
      - Update storage VM administrator account information (optional).
   b. For existing storage VMs:: click **Storage > Storage VMs**, select a storage VM, click **Settings**, and then click ⚙ under **SMB**.
2. Open the export policy of the storage VM root volume:
   a. Click **Storage > Volumes**, select the root volume of the storage VM (which by default is *volume-name_root*), and then click on the policy that is displayed under **Export Policy**.
   b. Click **Add** to add a rule.
      - Client specification = `0.0.0.0/0`
      - Access protocols = SMB
      - Access details = NTFS Read-Only
3. Configure DNS for host-name resolution:
   a. Click **Storage > Storage VMs**, select the storage VM, click **Settings**, and then click ⚙ under **DNS**.
   b. Switch to the DNS server and map the SMB server.
      - Create forward (A - Address record) and reverse (PTR - Pointer record) lookup entries to map the SMB server name to the IP address of the data network interface.
      - If you use NetBIOS aliases, create an alias canonical name (CNAME resource record) lookup entry to map each alias to the IP address of the SMB server's data network interface.

4. Configure name services as required
   a. Click **Storage > Storage VMs**, select the storage VM, click **Settings**, and then click ⚙ under **LDAP** or **NIS**.
   b. Include any changes in the name services switch file: click ✎ under **Name Services Switch**.
5. Configure Kerberos if required:
   a. Click **Storage > Storage VMs**, select the storage VM, and then click **Settings**.
   b. Click → under **Kerberos** and then click **Add**.

Enable NAS storage for both Windows and Linux using both NFS and SMB

Create or modify storage VMs to enable NFS and SMB servers to serve data to Linux and Windows clients.

This procedure enables a new or existing storage VM to serve both NFS and SMB protocols. It is assumed that configuration details are available for any networking, authentication, or security services required in your environment.



Steps

1. Enable NFS and SMB on a storage VM.
   a. For new storage VMs: click **Storage > Storage VMs**, click **Add**, enter a storage VM name, and in the **SMB/CIFS, NFS, S3** tab, select **Enable SMB/CIFS** and **Enable NFS**.
      - Enter the following information:
        - Administrator name and password
        - Server name
        - Active directory domain
      - Confirm the Organizational Unit.
      - Confirm the DNS values.
      - Confirm the default language.
      - Add network interfaces.
      - Update storage VM administrator account information (optional).
   b. For existing storage VMs: click **Storage > Storage VMs**, select a storage VM, and then click **Settings**. Complete the following sub-steps if NFS or SMB is not already enabled.
      - Click ⚙ under **NFS**.
      - Click ⚙ under **SMB**.
2. Open the export policy of the storage VM root volume:
   a. Click **Storage > Volumes**, select the root volume of the storage VM (which by default is *volume-name_root*), and then click on the policy that is displayed under **Export Policy**.

b. Click **Add** to add a rule.

   - Client specification = `0.0.0.0/0`
   - Access protocols = NFS
   - Access details = NFS Read-Only

3. Configure DNS for host-name resolution:

   a. Click **Storage > Storage VMs**, select the storage VM, click **Settings**, and then click ⚙ under **DNS**.

   b. When DNS configuration is complete, switch to the DNS server and map the SMB server.

      - Create forward (A - Address record) and reverse (PTR - Pointer record) lookup entries to map the SMB server name to the IP address of the data network interface.
      - If you use NetBIOS aliases, create an alias canonical name (CNAME resource record) lookup entry to map each alias to the IP address of the SMB server's data network interface.

4. Configure name services as required:

   a. Click **Storage > Storage VMs**, select the storage VM, click **Settings**, and then click ⚙ for LDAP or NIS.

   b. Include any changes in the name services switch file: click ✎ under **Name Services Switch**.

5. Configure Kerberos if required: click → in the Kerberos tile and then click **Add**.

6. Map UNIX and Windows user names if required: click → under **Name Mapping** and then click **Add**.

   You should use this procedure only if your site has Windows and UNIX user accounts that do not map implicitly, which is when the lowercase version of each Windows user name matches the UNIX user name. This procedure can be done using LDAP, NIS, or local users. If you have two sets of users that do not match, you should configure name mapping.

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